IGCSE 0607 Practice Questions — E1.3

Powers and roots

Total Questions: 18 — **Total Marks:** 94 Marks

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1 **(a)** Write down the value of $\sqrt{121}$.

..... **[1]**

(b) Work out the value of

$$4^3 - \sqrt[3]{27}$$

..... **[2]**

2 **(a)** Work out the value of

$$2^3 \times \sqrt{81}$$

..... **[2]**

(b) Write down the value of $\sqrt[3]{125}$.

..... **[1]**

3 **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

(b) Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

4 **(a)** Work out the value of

$3^3 - \sqrt{144}$

..... **[2]**

(b) Work out the value of

$2^4 \times \sqrt[3]{125}$

..... **[2]**

5 **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

(b) Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

(c) Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

6 **(a)** Work out the value of

$5^2 - \sqrt{121}$

..... **[1]**

(b) Work out the value of

$\sqrt[3]{27} \times 2^4$

..... **[2]**

(c) Work out the exact value of

$$\frac{6^2 + \sqrt{64}}{\sqrt[3]{125}}$$

..... **[2]**

7 **(a)** Work out the exact value of

$$\frac{5^2 + \sqrt{144}}{2^3 - \sqrt[3]{1}}$$

..... **[2]**

(b) Find the value of k when

$$3^k = \sqrt{81} \times 3^2$$

$k =$ ..... **[2]**

8 **(a)** Calculate the value of

$$\frac{4^3 + \sqrt{225}}{\sqrt[3]{64}}$$

Give your answer as a decimal.

..... **[3]**

(b) Given that $p = 5$ and $q = -2$, work out the value of

$$p^2 - q^3$$

..... **[2]**

9 **(a)** A cube has a volume of 125 cm^3.
Find the length of one edge of the cube.

..... cm **[1]**

(b) Work out the total surface area of this cube.

..... cm^2 **[2]**

(c) Given that $\sqrt{N} = 3^3 - 4^2$, find the value of N.

$N =$ ..... **[2]**

10 **(a)** Work out the exact value of

$$\left(\frac{\sqrt{225}+\sqrt[3]{1000}}{5^2}\right)^{-2}$$

..... **[3]**

(b) Solve the equation

$$x^{\frac{2}{3}} = \frac{\sqrt{144}}{3}$$

$x =$ ..... **[3]**

11 Solve the following equations.

(a)

$$\sqrt[3]{3x+4} = 4$$

$x =$ ..... **[3]**

(b)

$$x^2 = 4^3 + 6^2$$

$x =$ ..... , or $x =$ ..... **[2]**

(c)

$$\sqrt{x} + 2\sqrt{x} = 21$$

$x =$ ..... **[2]**

12 **(a)** Find the integer k such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

(b) Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

(c) Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of y^2.

..... **[3]**

13 **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

(c) Find the largest integer n such that $n^4 < 2000$.

$n =$ ..... **[2]**

14 **(a)** **(i)** Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

(b) Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**

15 **(a)** Calculate the value of

$$\frac{\sqrt{15.2^2 - 8.9^2}}{3.1^3}$$

..... **[2]**

(b) Given that $p = 4.2 \times 10^5$ and $q = 8.1 \times 10^{-3}$, work out the value of $\sqrt[3]{p \times q}$.
Give your answer correct to 3 significant figures.

..... **[3]**

16 **(a)** Calculate the value of

$$\frac{\sqrt[3]{45.6} + 2.1^4}{3.8^2}$$ Give your answer correct to 3 significant figures.

..... **[2]**

(b) A solid cube has a volume of 4913 cm^3.
Calculate the total surface area of the cube.

..... cm^2 **[3]**

17 **(a)** Calculate the value of

$\sqrt[5]{7.59375}$

..... **[1]**

(b) Find the two possible values of x in the equation

$2x^4 - 17 = 1233$

$x =$ ..... , $x =$ ..... **[3]**

(c) It is given that

$$y = \sqrt{\frac{3a^2 + b}{c}}$$

Work out the value of y when $a = 4.2$, $b = -5.7$ and $c = 1.8$.
Give your answer correct to 2 decimal places.

$y =$ ..... **[3]**

18 **(a)** Calculate the value of

$$\frac{\sqrt[4]{150000}}{4.5^3 - \sqrt{75}}$$ Give your answer correct to 3 decimal places.

..... **[2]**

(b) x is a positive integer such that $\sqrt[3]{x^2} = 16$.
Find the value of x.

$x =$ ..... **[3]**

(c) The volume of a sphere is equal to the volume of a cube with edge length 12 cm.
Calculate the radius of the sphere.
The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

..... cm **[3]**